

Assignment 16

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Solution: Task 1 - Load Test Using Locust:

1. Python script for the Locust test :

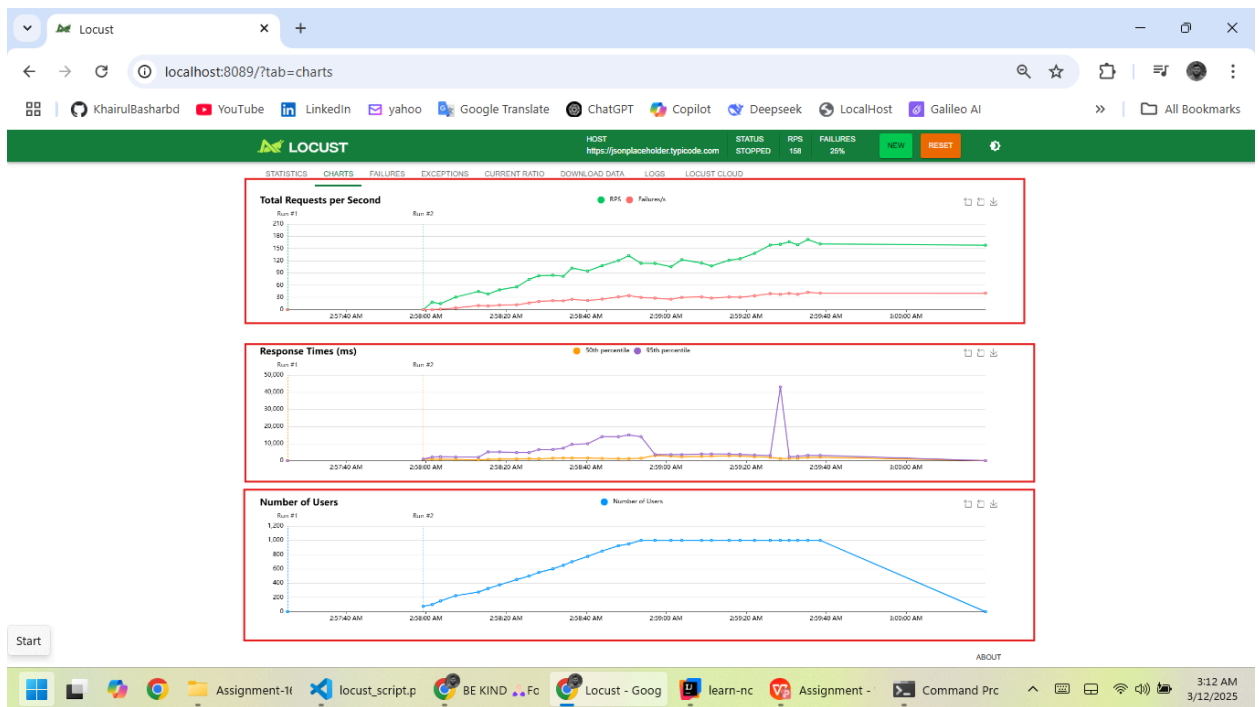
```
from locust import HttpUser, task, between

class WebsiteUser(HttpUser):
    wait_time = between(1, 7) # Simulate users waiting between 1 and 7
    seconds

    @task
    def load_homepage(self):
        self.client.get("/") # Test the homepage

    @task(3) # This task runs 3 times more often than the others
    def load_about_page(self):
        self.client.get("/posts") # Test the post page
```

2. Screenshot :



The screenshot shows the Locust web interface with the 'stats' tab selected. The interface displays a table with columns for Type, Name, # Requests, # Fails, Median (ms), 95%ile (ms), 99%ile (ms), Average (ms), Min (ms), Max (ms), Average size (bytes), Current RPS, and Current Failures/s. The table shows data for GET requests to '/' and '/posts', and an aggregated row. The interface also includes a 'Locust' logo and a 'stats' button.

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/	2846	2773	1700	4100	44000	2764.38	17	56093	5461.41	40.8	40.2
GET	/posts	8444	0	1700	4400	45000	2869.7	15	56091	27520	117.2	0
Aggregated		11290	2773	1700	4400	45000	2843.15	15	56093	21959.44	158	40.2

3. Summary of key findings from the load test:

I have setted total number of users 1000 and spawn rate 25 users per second

Load Test Summary

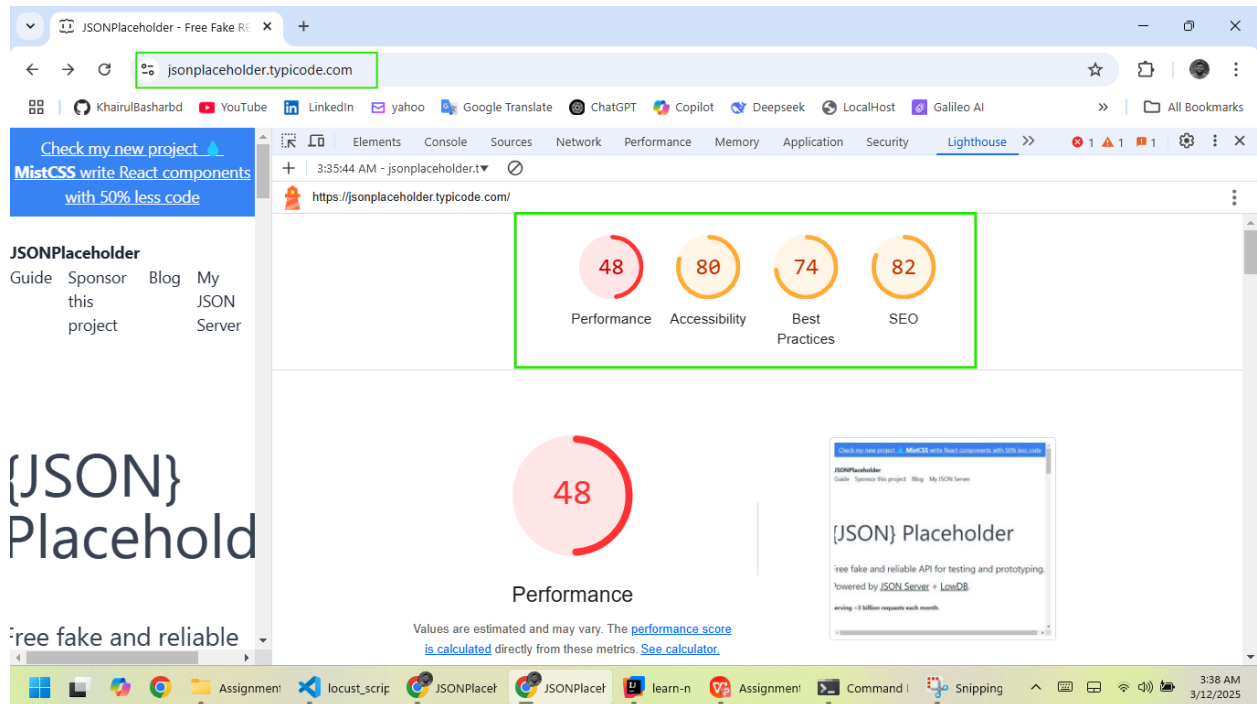
- **Total Users:** 1,000
- **Spawn Rate:** 25 users/second (40 seconds to full load)
- **Response Time:**
 - Average: 2800 ms
 - 95th Percentile: 4,000 ms
 - 99th Percentile: 4,5000 ms
- **Throughput:** 158 RPS (stable)
- **Failures:** 40 out of 11,000 requests

Key Insights:

- System handles 150 concurrent users well.
- Occasional bottlenecks (99th percentile response time).
- Low errors, but investigate root causes.

Solution: Task 2 - Google Lighthouse Performance Report

1. Screenshots:

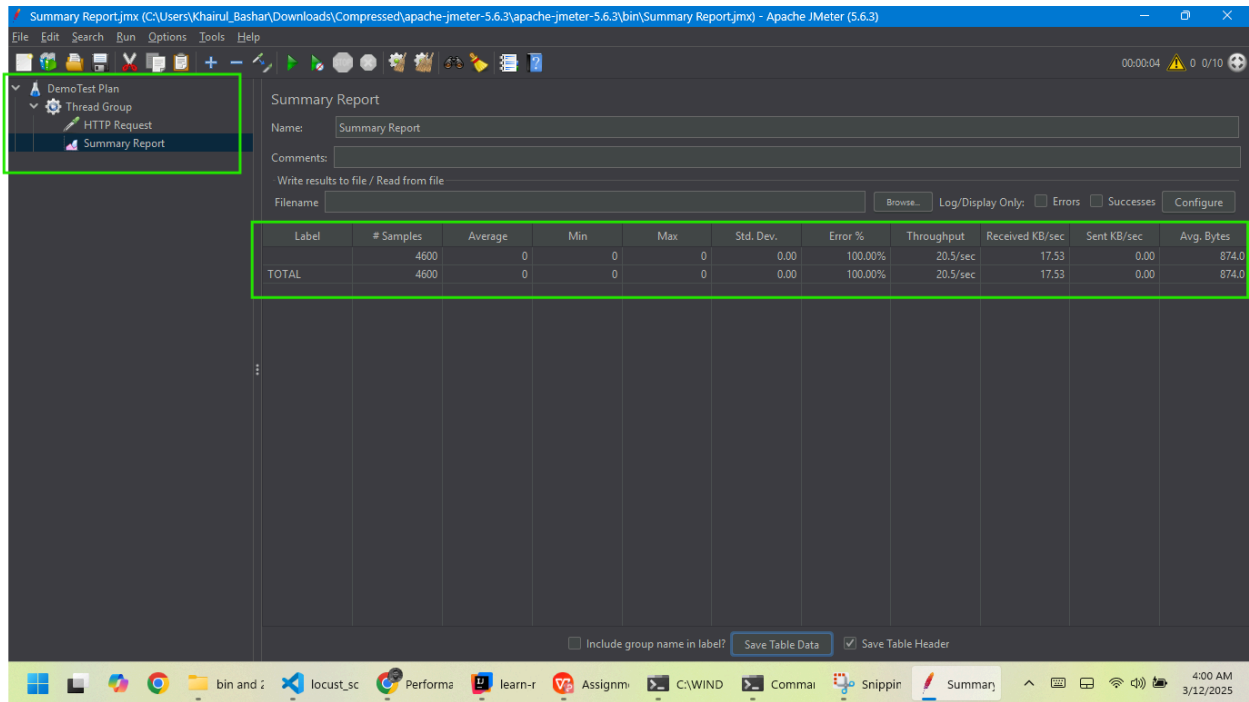


2. Keys Insights:

- a) **Performance (48 - Poor):** The website has a low performance score, indicating slow load times, inefficient rendering, or heavy resources.
- b) **Accessibility (80 - Good):** The site meets many accessibility standards but may need improvements for better usability.
- c) **Best Practices (74 - Needs Improvement):** Some coding standards and security measures might not be fully optimized.
- d) **SEO (82 - Fairly Good):** The site is reasonably optimized for search engines but could benefit from further improvements.

Solution: Task 3 - Load Testing Using JMeter:

Screenshots:



The screenshot displays the Apache JMeter 5.6.3 Summary Report window. The left sidebar shows the test plan structure: DemoTest Plan, Thread Group, HTTP Request, and Summary Report. The main area shows the Summary Report for the selected test plan. The report includes a table with the following data:

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
TOTAL	4600	0	0	0	0.00	100.00%	20.5/sec	17.53	0.00	874.0

The table is highlighted with a green border. Below the table, there are checkboxes for 'Include group name in label?' and 'Save Table Data', and a 'Save Table Header' checkbox. The status bar at the bottom shows the time as 4:00 AM on 3/12/2025.